



Mathematics: AI 1.1 Number systems, rounding and estimation	Mathematics: AI 1.1 Number systems, rounding and estimation	Mathematics: AI 1.1 Number systems, rounding and estimation
What distinguishes a rational number from an irrational number?	What is the formula for percentage error, and why is the result always positive?	Which of the following is a rational number? A 0.101001000100001... B $\sqrt[3]{5}$ C 2π D $-0.6666\ldots$ (recurring)
IB-MATHAI-1.1-1 CONCEPT	IB-MATHAI-1.1-2 CONCEPT	IB-MATHAI-1.1-3 MULTIPLE CHOICE
Mathematics: AI 1.1 Number systems, rounding and estimation	Mathematics: AI 1.2 Sequences and series	Mathematics: AI 1.2 Sequences and series
Round 0.049826 to 2 significant figures.	State the formula for the sum of the first n terms of a geometric sequence with first term u_1 and common ratio $r \neq 1$.	Under what condition does an infinite geometric series converge, and what is its sum?
IB-MATHAI-1.1-4 PRACTICE	IB-MATHAI-1.2-1 CONCEPT	IB-MATHAI-1.2-2 CONCEPT
Mathematics: AI 1.2 Sequences and series	Mathematics: AI 1.2 Sequences and series	Mathematics: AI 1.3 Exponents and logarithms
The 3rd term of an arithmetic sequence is 11 and the 7th term is 27. What is the common difference? A 3 B 4 C 5 D 16	A geometric sequence has $u_1=5$ and $r=2$. Find S_6 .	State the definition linking logarithms and exponents.
IB-MATHAI-1.2-3 MULTIPLE CHOICE	IB-MATHAI-1.2-4 PRACTICE	IB-MATHAI-1.3-1 CONCEPT
Mathematics: AI 1.3 Exponents and logarithms	Mathematics: AI 1.3 Exponents and logarithms	Mathematics: AI 1.3 Exponents and logarithms
What is the change of base formula, and why is it useful?	If $\log_5(x)=3$, what is x? A 15 B 25 C 125 D 243	Solve $2^x=7$ for x, correct to 3 significant figures.
IB-MATHAI-1.3-2 CONCEPT	IB-MATHAI-1.3-3 MULTIPLE CHOICE	IB-MATHAI-1.3-4 PRACTICE
Mathematics: AI 1.4 Financial mathematics: loans and amortization	Mathematics: AI 1.4 Financial mathematics: loans and amortization	Mathematics: AI 1.4 Financial mathematics: loans and amortization
In loan amortization, why does the portion of each fixed payment going towards interest decrease over time?	What five values are used in a TVM/finance solver, and what does each represent?	A loan is fully amortized over its term. What is the future value (FV) at the end of the term? A Equal to PV B Equal to total interest paid C Zero D Equal to one payment
IB-MATHAI-1.4-1 CONCEPT	IB-MATHAI-1.4-2 CONCEPT	IB-MATHAI-1.4-3 MULTIPLE CHOICE
Mathematics: AI 1.4 Financial mathematics: loans and amortization	Mathematics: AI 1.5 Matrices and systems of equations (HL)	Mathematics: AI 1.5 Matrices and systems of equations (HL)
A \$5000 loan at 12% p.a. compounded monthly is repaid over 12 months. Find the monthly interest rate used in the amortization formula.	How do you find the determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$?	What does $\det(A)=0$ tell you about the system of equations $Ax=b$?
IB-MATHAI-1.4-4 PRACTICE	IB-MATHAI-1.5-1 CONCEPT	IB-MATHAI-1.5-2 CONCEPT



Mathematics: AI 1.5 Matrices and systems of equations (HL) For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, what is $\det(A)$? A -2 B 2 C -10 D 10 IB-MATHAI-1.5-3 MULTIPLE CHOICE	Mathematics: AI 1.5 Matrices and systems of equations (HL) Find the inverse of $A = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$. IB-MATHAI-1.5-4 PRACTICE	Mathematics: AI 1.6 Eigenvalues and eigenvectors (HL) How do you find the eigenvalues of a square matrix A? IB-MATHAI-1.6-1 CONCEPT
Mathematics: AI 1.6 Eigenvalues and eigenvectors (HL) Once an eigenvalue λ is found, how do you find its corresponding eigenvector? IB-MATHAI-1.6-2 CONCEPT	Mathematics: AI 1.6 Eigenvalues and eigenvectors (HL) A 2×2 matrix has eigenvalues 3 and -1 . Which statement is necessarily true? A The matrix is singular B The matrix has determinant -3 C The matrix has determinant 3 D The trace is -3 IB-MATHAI-1.6-3 MULTIPLE CHOICE	Mathematics: AI 1.6 Eigenvalues and eigenvectors (HL) Matrix $A = \begin{bmatrix} 5 & 0 \\ 0 & -2 \end{bmatrix}$ is diagonal. State its eigenvalues and corresponding eigenvectors. IB-MATHAI-1.6-4 PRACTICE
Mathematics: AI 2.1 Linear models In a linear model $y = mx + c$ applied to a real context, what do m and c represent? IB-MATHAI-2.1-1 CONCEPT	Mathematics: AI 2.1 Linear models What is the danger of using a linear model to extrapolate far beyond the range of the original data? IB-MATHAI-2.1-2 CONCEPT	Mathematics: AI 2.1 Linear models A linear model has equation $P = 120 - 4t$. What is the rate of change of P with respect to t? A 120 B -4 C 4 D -120 IB-MATHAI-2.1-3 MULTIPLE CHOICE
Mathematics: AI 2.1 Linear models A gym charges a \$20 joining fee plus \$15 per month. Write a linear model for total cost C after m months, and find C after 6 months. IB-MATHAI-2.1-4 PRACTICE	Mathematics: AI 2.2 Quadratic models How do you find the x-coordinate of the vertex of a parabola $y = ax^2 + bx + c$? IB-MATHAI-2.2-1 CONCEPT	Mathematics: AI 2.2 Quadratic models If a quadratic model has a negative leading coefficient ($a < 0$), what does this tell you about its vertex? IB-MATHAI-2.2-2 CONCEPT
Mathematics: AI 2.2 Quadratic models A parabola has equation $y = -(x - 4)^2 + 9$. What is its maximum value? A 4 B -4 C 9 D -9 IB-MATHAI-2.2-3 MULTIPLE CHOICE	Mathematics: AI 2.2 Quadratic models Find the roots of $y = x^2 - 5x + 6$ by factorisation. IB-MATHAI-2.2-4 PRACTICE	Mathematics: AI 2.3 Exponential and logarithmic models In the model $N(t) = N_0 e^{kt}$, what does the sign of k tell you? IB-MATHAI-2.3-1 CONCEPT
Mathematics: AI 2.3 Exponential and logarithmic models How do you find the half-life of a quantity decaying as $N(t) = N_0 e^{kt}$ with $k < 0$? IB-MATHAI-2.3-2 CONCEPT	Mathematics: AI 2.3 Exponential and logarithmic models A quantity grows according to $V(t) = 80(1.06)^t$. What is the annual percentage growth rate? A 1.06% B 6% C 60% D 106% IB-MATHAI-2.3-3 MULTIPLE CHOICE	Mathematics: AI 2.3 Exponential and logarithmic models A car worth \$24000 depreciates so that its value is $V(t) = 24000(0.9)^t$ after t years. Find its value after 3 years. IB-MATHAI-2.3-4 PRACTICE



Mathematics: AI 2.4 Cubic and other polynomial models	Mathematics: AI 2.4 Cubic and other polynomial models	Mathematics: AI 2.4 Cubic and other polynomial models
For a cubic $y=ax^3+bx^2+cx+d$ with $a>0$, describe the end behaviour as $x\rightarrow\pm\infty$.	Why might a cubic model be preferred over a quadratic model for a set of data?	A cubic model has three real roots: -2 , 1 , and 4 . Which is a correct factorised form? A $y=a(x+2)(x-1)(x-4)$ B $y=a(x-2)(x+1)(x+4)$ C $y=a(x+2)(x-1)(x+4)$ D $y=a(x-2)(x-1)(x-4)$
IB-MATHAI-2.4-1 CONCEPT	IB-MATHAI-2.4-2 CONCEPT	IB-MATHAI-2.4-3 MULTIPLE CHOICE
Mathematics: AI 2.4 Cubic and other polynomial models	Mathematics: AI 2.5 Modelling with technology and model fitting	Mathematics: AI 2.5 Modelling with technology and model fitting
A cubic volume model gives $V(2)=40$ for $V(x)$, where x is a side length in cm. State what this value represents.	What does the coefficient of determination R^2 measure when fitting a regression model?	Besides checking R^2 , what other consideration should guide the choice of regression model for real data?
IB-MATHAI-2.4-4 PRACTICE	IB-MATHAI-2.5-1 CONCEPT	IB-MATHAI-2.5-2 CONCEPT
Mathematics: AI 2.5 Modelling with technology and model fitting	Mathematics: AI 2.5 Modelling with technology and model fitting	Mathematics: AI 2.6 Optimisation of models (HL)
Two models are fitted to the same data: Model A has $R^2=0.72$, Model B has $R^2=0.91$. Based on R^2 alone, which fits better? A Model A B Model B C Both fit equally well D Cannot be determined	A linear regression on a data set gives $R^2=0.36$. What does this suggest about the linear model's suitability?	In a one-variable optimisation problem, what two types of point must be checked to find the global maximum or minimum of $f(x)$ on a closed domain?
IB-MATHAI-2.5-3 MULTIPLE CHOICE	IB-MATHAI-2.5-4 PRACTICE	IB-MATHAI-2.6-1 CONCEPT
Mathematics: AI 2.6 Optimisation of models (HL)	Mathematics: AI 2.6 Optimisation of models (HL)	Mathematics: AI 2.6 Optimisation of models (HL)
Why is a constraint equation (e.g. fixed volume or perimeter) useful in a real-world optimisation problem?	$f(x)=x^3-3x$ has a stationary point at $x=1$, and $f''(x)=6x$. What type of stationary point is $x=1$? A Local maximum B Local minimum C Point of inflection D Cannot be determined	A rectangle has perimeter 40 cm. Express its area A in terms of one side length x , and state the domain of x .
IB-MATHAI-2.6-2 CONCEPT	IB-MATHAI-2.6-3 MULTIPLE CHOICE	IB-MATHAI-2.6-4 PRACTICE
Mathematics: AI 3.1 Coordinate geometry and Voronoi diagrams	Mathematics: AI 3.1 Coordinate geometry and Voronoi diagrams	Mathematics: AI 3.1 Coordinate geometry and Voronoi diagrams
What is the perpendicular bisector of a line segment, and what key property does every point on it have?	What defines the point where three regions meet at a vertex in a Voronoi diagram?	What is the midpoint of the segment joining $(-2,5)$ and $(4,-1)$? A $(1,2)$ B $(2,1)$ C $(1,3)$ D $(3,1)$
IB-MATHAI-3.1-1 CONCEPT	IB-MATHAI-3.1-2 CONCEPT	IB-MATHAI-3.1-3 MULTIPLE CHOICE
Mathematics: AI 3.1 Coordinate geometry and Voronoi diagrams	Mathematics: AI 3.2 Trigonometry: non-right triangles	Mathematics: AI 3.2 Trigonometry: non-right triangles
Find the distance between points $P(1,2)$ and $Q(4,6)$.	State the cosine rule for finding side a , given sides b , c and included angle A .	When using the sine rule to find an angle from SSA information, why might there be two possible answers (the ambiguous case)?
IB-MATHAI-3.1-4 PRACTICE	IB-MATHAI-3.2-1 CONCEPT	IB-MATHAI-3.2-2 CONCEPT



Mathematics: AI 3.2 Trigonometry: non-right triangles	Mathematics: AI 3.2 Trigonometry: non-right triangles	Mathematics: AI 3.3 Volume, surface area and geometric applications
Triangle ABC has $b=8$, $c=10$, angle $A=60^\circ$. What is a^2 ? A 36 B 84 C 44 D 164	Find the area of a triangle with sides $a=6$, $b=9$ and included angle $C=40^\circ$.	State the formula for the volume of a cone with radius r and height h .
IB-MATHAI-3.2-3 MULTIPLE CHOICE	IB-MATHAI-3.2-4 PRACTICE	IB-MATHAI-3.3-1 CONCEPT
Mathematics: AI 3.3 Volume, surface area and geometric applications	Mathematics: AI 3.3 Volume, surface area and geometric applications	Mathematics: AI 3.3 Volume, surface area and geometric applications
When a composite solid is formed by removing a smaller solid from a larger one, how is its volume found?	A sphere has radius 6 cm. What is its surface area? A $36\pi \text{ cm}^2$ B $144\pi \text{ cm}^2$ C $288\pi \text{ cm}^2$ D $72\pi \text{ cm}^2$	Find the volume of a cylinder with radius 2 cm and height 9 cm, in terms of π .
IB-MATHAI-3.3-2 CONCEPT	IB-MATHAI-3.3-3 MULTIPLE CHOICE	IB-MATHAI-3.3-4 PRACTICE
Mathematics: AI 3.4 Vectors (HL)	Mathematics: AI 3.4 Vectors (HL)	Mathematics: AI 3.4 Vectors (HL)
How is the magnitude of a 3D vector $v=(x,y,z)$ calculated?	What does it mean if the scalar (dot) product of two non-zero vectors equals zero?	Given $a=(1,0,2)$ and $b=(2,1,-1)$, find $a+2b$. A (5,2,0) B (3,1,1) C (5,2,-4) D (4,1,0)
IB-MATHAI-3.4-1 CONCEPT	IB-MATHAI-3.4-2 CONCEPT	IB-MATHAI-3.4-3 MULTIPLE CHOICE
Mathematics: AI 3.4 Vectors (HL)	Mathematics: AI 3.5 Vector applications to kinematics (HL)	Mathematics: AI 3.5 Vector applications to kinematics (HL)
Find the magnitude of vector $v=(6,-8,0)$.	If a particle's position vector is $r(t)$, how do you find its velocity and acceleration vectors?	How is the speed of a particle found from its velocity vector $v(t)$?
IB-MATHAI-3.4-4 PRACTICE	IB-MATHAI-3.5-1 CONCEPT	IB-MATHAI-3.5-2 CONCEPT
Mathematics: AI 3.5 Vector applications to kinematics (HL)	Mathematics: AI 3.5 Vector applications to kinematics (HL)	Mathematics: AI 4.1 Sampling techniques
A particle has position vector $r(t)=(t^2,3t)$. What is its velocity vector at $t=2$? A (2,3) B (4,3) C (4,6) D (2,6)	Two particles have position vectors $r_A(t)=(t,2t)$ and $r_B(t)=(4-t,1)$. Find the value of t at which their x-coordinates are equal.	In stratified sampling, how is the sample size allocated to each stratum?
IB-MATHAI-3.5-3 MULTIPLE CHOICE	IB-MATHAI-3.5-4 PRACTICE	IB-MATHAI-4.1-1 CONCEPT
Mathematics: AI 4.1 Sampling techniques	Mathematics: AI 4.1 Sampling techniques	Mathematics: AI 4.1 Sampling techniques
What is a key disadvantage of convenience sampling?	A population of 500 is divided into two strata of 300 and 200. For a stratified sample of size 40, how many should come from the stratum of 200? A 12 B 16 C 20 D 24	A list of 1000 customers is numbered 1–1000. Every 20th customer is selected, starting from a random number between 1 and 20. What sampling method is this?
IB-MATHAI-4.1-2 CONCEPT	IB-MATHAI-4.1-3 MULTIPLE CHOICE	IB-MATHAI-4.1-4 PRACTICE



Mathematics: AI 4.2 Summarising data: measures and displays	Mathematics: AI 4.2 Summarising data: measures and displays	Mathematics: AI 4.2 Summarising data: measures and displays
Why is the median often preferred over the mean as a measure of central tendency when a data set contains extreme outliers?	How is the interquartile range (IQR) calculated, and what does it measure?	A box-and-whisker plot shows $Q1=5$, median= 9 , $Q3=14$. What is the interquartile range? A 4 B 5 C 9 D 14
IB-MATHAI-4.2-1 CONCEPT	IB-MATHAI-4.2-2 CONCEPT	IB-MATHAI-4.2-3 MULTIPLE CHOICE
Mathematics: AI 4.2 Summarising data: measures and displays	Mathematics: AI 4.3 Bivariate data and correlation	Mathematics: AI 4.3 Bivariate data and correlation
A data set is 4,7,7,9,10,13,15. Find the median and the range.	What range of values can Pearson's correlation coefficient r take, and what do values near $+1$, -1 and 0 indicate?	Why is 'correlation does not imply causation' an important principle when interpreting a strong r value?
IB-MATHAI-4.2-4 PRACTICE	IB-MATHAI-4.3-1 CONCEPT	IB-MATHAI-4.3-2 CONCEPT
Mathematics: AI 4.3 Bivariate data and correlation	Mathematics: AI 4.3 Bivariate data and correlation	Mathematics: AI 4.4 Probability, Venn and tree diagrams
A regression line $y=3x+10$ is valid for x between 0 and 20 in the original data. Using it to predict y at $x=50$ is an example of: A Interpolation B Extrapolation C Correlation D Causation	A data set gives correlation coefficient $r=0.65$. Describe the strength and direction of the linear correlation.	State the general addition rule for $P(A \cup B)$ for two events A and B .
IB-MATHAI-4.3-3 MULTIPLE CHOICE	IB-MATHAI-4.3-4 PRACTICE	IB-MATHAI-4.4-1 CONCEPT
Mathematics: AI 4.4 Probability, Venn and tree diagrams	Mathematics: AI 4.4 Probability, Venn and tree diagrams	Mathematics: AI 4.4 Probability, Venn and tree diagrams
How does the formula for $P(A \cap B)$ differ between independent and dependent events?	Events A and B are mutually exclusive with $P(A)=0.3$ and $P(B)=0.45$. Find $P(A \cup B)$. A 0.135 B 0.75 C 0.15 D 1	A die is rolled once. Find the probability of rolling a number greater than 4.
IB-MATHAI-4.4-2 CONCEPT	IB-MATHAI-4.4-3 MULTIPLE CHOICE	IB-MATHAI-4.4-4 PRACTICE
Mathematics: AI 4.5 Discrete distributions and the binomial distribution	Mathematics: AI 4.5 Discrete distributions and the binomial distribution	Mathematics: AI 4.5 Discrete distributions and the binomial distribution
State the conditions required for a random variable X to follow a binomial distribution $B(n,p)$.	State the formulas for the mean and variance of $X \sim B(n,p)$.	$X \sim B(8,0.3)$. What is $\text{Var}(X)$? A 2.4 B 1.68 C 0.24 D 5.6
IB-MATHAI-4.5-1 CONCEPT	IB-MATHAI-4.5-2 CONCEPT	IB-MATHAI-4.5-3 MULTIPLE CHOICE
Mathematics: AI 4.5 Discrete distributions and the binomial distribution	Mathematics: AI 4.6 The normal distribution	Mathematics: AI 4.6 The normal distribution
$X \sim B(5,0.4)$. Find $P(X=0)$.	For a normal distribution, approximately what percentage of values lie within 2 standard deviations of the mean?	How is a z-score calculated, and what does it represent?
IB-MATHAI-4.5-4 PRACTICE	IB-MATHAI-4.6-1 CONCEPT	IB-MATHAI-4.6-2 CONCEPT



Mathematics: AI 4.6 The normal distribution	Mathematics: AI 4.6 The normal distribution	Mathematics: AI 4.7 The chi-squared test
$X \sim N(60, 25)$. Find the z-score for $x = 70$. A 0.4 B 2 C 10 D 1.4	$X \sim N(30, 4^2)$. Find $P(X < 30)$, using the symmetry of the normal distribution.	State the formula for the chi-squared test statistic and for expected frequency in a contingency table.
IB-MATHAI-4.6-3MULTIPLE CHOICE	IB-MATHAI-4.6-4PRACTICE	IB-MATHAI-4.7-1CONCEPT
Mathematics: AI 4.7 The chi-squared test	Mathematics: AI 4.7 The chi-squared test	Mathematics: AI 4.7 The chi-squared test
How is the decision to reject or not reject H_0 made in a chi-squared test using the critical value approach?	A chi-squared goodness-of-fit test compares observed frequencies to: A Frequencies from a different sample B Frequencies expected under a hypothesised distribution C The sample mean D The sample median	In a 2×2 contingency table, what are the degrees of freedom for the chi-squared test of independence?
IB-MATHAI-4.7-2CONCEPT	IB-MATHAI-4.7-3MULTIPLE CHOICE	IB-MATHAI-4.7-4PRACTICE
Mathematics: AI 4.8 Spearman's rank correlation	Mathematics: AI 4.8 Spearman's rank correlation	Mathematics: AI 4.8 Spearman's rank correlation
State the formula for Spearman's rank correlation coefficient r_s in terms of the rank differences d and number of pairs n.	Why might Spearman's rank correlation be used instead of Pearson's correlation coefficient?	If Spearman's rank correlation coefficient $r_s = -0.9$, what does this indicate? A A weak monotonic relationship B A strong, mostly decreasing monotonic relationship C No relationship D A strong increasing linear relationship only
IB-MATHAI-4.8-1CONCEPT	IB-MATHAI-4.8-2CONCEPT	IB-MATHAI-4.8-3MULTIPLE CHOICE
Mathematics: AI 4.8 Spearman's rank correlation	Mathematics: AI 4.9 Further hypothesis tests: the t-test (HL)	Mathematics: AI 4.9 Further hypothesis tests: the t-test (HL)
Two rankings of 4 items have rank differences $d = 1, -1, 1, -1$. Find $\sum d^2$ and hence r_s using $n = 4$.	What is the key difference in when to use a t-test versus a z-test for a hypothesis test about a mean?	In hypothesis testing, what is the decision rule based on the p-value and significance level α?
IB-MATHAI-4.8-4PRACTICE	IB-MATHAI-4.9-1CONCEPT	IB-MATHAI-4.9-2CONCEPT
Mathematics: AI 4.9 Further hypothesis tests: the t-test (HL)	Mathematics: AI 4.9 Further hypothesis tests: the t-test (HL)	Mathematics: AI 4.10 Non-linear regression and the Poisson distribution (HL)
A two-sample t-test is used to: A Test a single sample mean against a known value B Compare the means of two independent samples C Test for association between two categorical variables D Test independence in a contingency table	A one-sample t-test gives a p-value of 0.08 at a 5% significance level. What is the conclusion?	State the probability mass function for a Poisson random variable $X \sim \text{Po}(m)$, and its mean and variance.
IB-MATHAI-4.9-3MULTIPLE CHOICE	IB-MATHAI-4.9-4PRACTICE	IB-MATHAI-4.10-1CONCEPT



Mathematics: AI 4.10 Non-linear regression and the Poisson distribution (HL) Under what conditions is the Poisson distribution an appropriate model for a random variable? IB-MATHAI-4.10-2 CONCEPT	Mathematics: AI 4.10 Non-linear regression and the Poisson distribution (HL) If $X \sim \text{Po}(2)$, find $P(X=1)$. A e^{-2} B $2e^{-2}$ C $4e^{-2}$ D $0.5e^{-2}$ IB-MATHAI-4.10-3 MULTIPLE CHOICE	Mathematics: AI 4.10 Non-linear regression and the Poisson distribution (HL) A power model $y=3x^{0.5}$ is fitted to data with $R^2=0.99$. What does this suggest about the model? IB-MATHAI-4.10-4 PRACTICE
Mathematics: AI 5.1 Rates of change and the derivative What does the derivative $f'(a)$ represent geometrically and physically? IB-MATHAI-5.1-1 CONCEPT	Mathematics: AI 5.1 Rates of change and the derivative State the derivative of $e^{(kx)}$ and of $\ln(x)$ with respect to x . IB-MATHAI-5.1-2 CONCEPT	Mathematics: AI 5.1 Rates of change and the derivative Find dy/dx for $y=3\sin(x)+\cos(x)$. A $3\cos(x)-\sin(x)$ B $3\cos(x)+\sin(x)$ C $-3\sin(x)+\cos(x)$ D $-3\cos(x)-\sin(x)$ IB-MATHAI-5.1-3 MULTIPLE CHOICE
Mathematics: AI 5.1 Rates of change and the derivative Find the gradient of $y=x^2-4x+3$ at $x=1$. IB-MATHAI-5.1-4 PRACTICE	Mathematics: AI 5.2 Optimisation with technology In an optimisation problem, after finding a candidate value from $f'(x)=0$, what should always be checked? IB-MATHAI-5.2-1 CONCEPT	Mathematics: AI 5.2 Optimisation with technology Why is a constraint equation (like a fixed volume or fixed perimeter) important in a real-world optimisation problem? IB-MATHAI-5.2-2 CONCEPT
Mathematics: AI 5.2 Optimisation with technology A surface area function $S(r)$ has $S'(r)=0$ at $r=3$, with $S''(3)>0$. What does this tell you? A $r=3$ gives a local maximum B $r=3$ gives a local minimum C $r=3$ is not a stationary point D $S(3)=0$ IB-MATHAI-5.2-3 MULTIPLE CHOICE	Mathematics: AI 5.2 Optimisation with technology A rectangle's area is $A(x)=x(30-x)$ for $0<x<30$. Find the value of x that maximizes the area. IB-MATHAI-5.2-4 PRACTICE	Mathematics: AI 5.3 Integration and the trapezoidal rule State the trapezoidal rule formula for approximating $\int[a,b] f(x)dx$ using n equal intervals of width $h=(b-a)/n$, with y -values $y_0, \dots, y_n$. IB-MATHAI-5.3-1 CONCEPT
Mathematics: AI 5.3 Integration and the trapezoidal rule How does the trapezoidal rule's estimate compare to the true value of a definite integral when the function is concave down on the interval? IB-MATHAI-5.3-2 CONCEPT	Mathematics: AI 5.3 Integration and the trapezoidal rule $\int(4x^3-2x) dx = ?$ A x^4-x^2+c B $4x^4-2x^2+c$ C x^4-2x^2+c D $12x^2-2+c$ IB-MATHAI-5.3-3 MULTIPLE CHOICE	Mathematics: AI 5.3 Integration and the trapezoidal rule Estimate $\int[0,2] f(x)dx$ using the trapezoidal rule with 2 intervals, given $f(0)=3$, $f(1)=5$, $f(2)=4$. IB-MATHAI-5.3-4 PRACTICE
Mathematics: AI 5.4 Kinematics using calculus For a particle with displacement $s(t)$, state how velocity and acceleration are obtained, and how displacement is recovered from velocity. IB-MATHAI-5.4-1 CONCEPT	Mathematics: AI 5.4 Kinematics using calculus What is the key difference between 'displacement' and 'total distance travelled' when velocity changes sign over an interval? IB-MATHAI-5.4-2 CONCEPT	Mathematics: AI 5.4 Kinematics using calculus A particle has displacement $s(t)=t^3-3t$. What is its acceleration function $a(t)$? A $3t^2-3$ B $6t$ C t^2-3 D $3t^2$ IB-MATHAI-5.4-3 MULTIPLE CHOICE

**Mathematics: AI**

5.4 Kinematics using calculus

A particle has constant velocity $v(t)=5$ m/s for $0 \leq t \leq 4$ s. Find the distance travelled.

IB-MATHAI-5.4-4

PRACTICE

Mathematics: AI

5.5 Differential equations and Euler's method (HL)

State Euler's method's iterative formulas for approximating a solution to $dy/dx=f(x,y)$ with initial condition (x_0,y_0) and step size h .

IB-MATHAI-5.5-1

CONCEPT

Mathematics: AI

5.5 Differential equations and Euler's method (HL)

What happens to the accuracy of Euler's method as the step size h is decreased?

IB-MATHAI-5.5-2

CONCEPT

Mathematics: AI

5.5 Differential equations and Euler's method (HL)

Given $dy/dx=2x$, $y(0)=0$, using Euler's method with $h=1$, find $y(1)$.

- A 0
B 1
C 2
D 4

IB-MATHAI-5.5-3

MULTIPLE CHOICE

Mathematics: AI

5.5 Differential equations and Euler's method (HL)

For $dy/dx=y$, $y(0)=1$, use one Euler step with $h=0.1$ to estimate $y(0.1)$.

IB-MATHAI-5.5-4

PRACTICE

Mathematics: AI

5.6 Further calculus applications to modelling (HL)

State the chain rule as used in related rates problems, connecting dA/dt , dA/dr and dr/dt .

IB-MATHAI-5.6-1

CONCEPT

Mathematics: AI

5.6 Further calculus applications to modelling (HL)

How is the area between two curves $y=f(x)$ and $y=g(x)$ (with $f(x) \geq g(x)$) found on an interval $[a,b]$?

IB-MATHAI-5.6-2

CONCEPT

Mathematics: AI

5.6 Further calculus applications to modelling (HL)

A cube's volume $V=s^3$ (s =side length) is increasing with $dV/dt=12$ cm^3/s . What is dV/ds ?

- A $3s$
B s^2
C $3s^2$
D s^3

IB-MATHAI-5.6-3

MULTIPLE CHOICE

Mathematics: AI

5.6 Further calculus applications to modelling (HL)

Find the area enclosed between $y=x^2$ and $y=4$ for x between -2 and 2 .

IB-MATHAI-5.6-4

PRACTICE



Mathematics: AI 1.1 Number systems, rounding and estimation	Mathematics: AI 1.1 Number systems, rounding and estimation	Mathematics: AI 1.1 Number systems, rounding and estimation
D. A recurring decimal always equals a fraction; here $-0.6666... = -2/3$.	Percentage error = $ \text{approximate value} - \text{exact value} / \text{exact value} \times 100\%$. The absolute value is used because error measures the size of the deviation, not its direction.	A rational number can be written as p/q with integers p, q ($q \neq 0$) and has a terminating or repeating decimal expansion; an irrational number cannot be written this way and has a non-terminating, non-repeating decimal expansion (e.g. $\sqrt{2}$, π).
IB-MATHAI-1.1-3ANSWER	IB-MATHAI-1.1-2ANSWER	IB-MATHAI-1.1-1ANSWER
Mathematics: AI 1.2 Sequences and series	Mathematics: AI 1.2 Sequences and series	Mathematics: AI 1.1 Number systems, rounding and estimation
It converges when $ r < 1$; the sum is $S_{\infty} = u_1/(1-r)$.	$S_n = u_1(r^n - 1)/(r - 1)$, equivalently $u_1(1 - r^n)/(1 - r)$.	The first two significant figures are 4 and 9; the next digit (8) rounds the 9 up, carrying over, giving 0.050 (2 sf).
IB-MATHAI-1.2-2ANSWER	IB-MATHAI-1.2-1ANSWER	IB-MATHAI-1.1-4ANSWER
Mathematics: AI 1.3 Exponents and logarithms	Mathematics: AI 1.2 Sequences and series	Mathematics: AI 1.2 Sequences and series
$\log_a(x) = y$ is equivalent to $a^y = x$, for $a > 0$, $a \neq 1$, $x > 0$.	$S_6 = 5(2^6 - 1)/(2 - 1) = 5 \times 63 = 315$.	B. $d = (27 - 11)/(7 - 3) = 16/4 = 4$.
IB-MATHAI-1.3-1ANSWER	IB-MATHAI-1.2-4ANSWER	IB-MATHAI-1.2-3ANSWER
Mathematics: AI 1.3 Exponents and logarithms	Mathematics: AI 1.3 Exponents and logarithms	Mathematics: AI 1.3 Exponents and logarithms
$x = \ln 7 / \ln 2 = 1.9459 / 0.6931 \approx 2.81$.	C. $5^3 = 125$.	$\log_a(b) = \ln(b)/\ln(a)$; it lets any logarithm be evaluated on a calculator, since calculators only compute natural or base-10 logs directly.
IB-MATHAI-1.3-4ANSWER	IB-MATHAI-1.3-3ANSWER	IB-MATHAI-1.3-2ANSWER
Mathematics: AI 1.4 Financial mathematics: loans and amortization	Mathematics: AI 1.4 Financial mathematics: loans and amortization	Mathematics: AI 1.4 Financial mathematics: loans and amortization
C. A fully amortized loan has zero balance remaining after the final payment.	N (number of payments), I% (annual interest rate), PV (present/loan value), PMT (regular payment), and FV (future/remaining value, usually 0 for a fully repaid loan). Given any four, the solver finds the fifth.	Each payment first covers interest on the current outstanding balance; as the balance falls after each payment, the interest owed shrinks, so a growing share of the fixed payment reduces the principal instead.
IB-MATHAI-1.4-3ANSWER	IB-MATHAI-1.4-2ANSWER	IB-MATHAI-1.4-1ANSWER
Mathematics: AI 1.5 Matrices and systems of equations (HL)	Mathematics: AI 1.5 Matrices and systems of equations (HL)	Mathematics: AI 1.4 Financial mathematics: loans and amortization
A has no inverse, so the system does not have a unique solution — it has either no solutions or infinitely many, depending on b.	$\det = ad - bc$.	$i = 12\%/12 = 1\% = 0.01$ per month.
IB-MATHAI-1.5-2ANSWER	IB-MATHAI-1.5-1ANSWER	IB-MATHAI-1.4-4ANSWER
Mathematics: AI 1.6 Eigenvalues and eigenvectors (HL)	Mathematics: AI 1.5 Matrices and systems of equations (HL)	Mathematics: AI 1.5 Matrices and systems of equations (HL)
Solve the characteristic equation $\det(A - \lambda I) = 0$ for λ .	$\det = 10$; $A^{-1} = (1/10)[[5, 0], [0, 2]] = [[0.5, 0], [0, 0.2]]$.	A. $\det = 1 \times 4 - 2 \times 3 = 4 - 6 = -2$.
IB-MATHAI-1.6-1ANSWER	IB-MATHAI-1.5-4ANSWER	IB-MATHAI-1.5-3ANSWER



Mathematics: AI 1.6 Eigenvalues and eigenvectors (HL)	Mathematics: AI 1.6 Eigenvalues and eigenvectors (HL)	Mathematics: AI 1.6 Eigenvalues and eigenvectors (HL)
Eigenvalues are 5 and -2 (the diagonal entries); eigenvectors are $(1,0)$ for $\lambda=5$ and $(0,1)$ for $\lambda=-2$.	B. $\det(A)$ equals the product of the eigenvalues: $3 \times (-1) = -3$.	Substitute λ into $(A - \lambda I)v = 0$ and solve for a non-zero vector v ; the system is dependent, so solutions form a line and any non-zero scalar multiple is a valid eigenvector.
IB-MATHAI-1.6-4ANSWER	IB-MATHAI-1.6-3ANSWER	IB-MATHAI-1.6-2ANSWER
Mathematics: AI 2.1 Linear models	Mathematics: AI 2.1 Linear models	Mathematics: AI 2.1 Linear models
B. The gradient (coefficient of t) is -4 , meaning P decreases by 4 units per unit increase in t .	The relationship may not remain linear outside the observed range, so the further the extrapolation, the less reliable the prediction becomes.	m is the rate of change (gradient) — how much y changes per unit increase in x ; c is the value of y when $x=0$ (the fixed or starting value).
IB-MATHAI-2.1-3ANSWER	IB-MATHAI-2.1-2ANSWER	IB-MATHAI-2.1-1ANSWER
Mathematics: AI 2.2 Quadratic models	Mathematics: AI 2.2 Quadratic models	Mathematics: AI 2.1 Linear models
The parabola opens downward, so the vertex is a maximum point.	$x = -b/(2a)$; substitute this back into the equation to find the corresponding y -value (the maximum or minimum).	$C(m)=20+15m$; $C(6)=20+90=\$110$.
IB-MATHAI-2.2-2ANSWER	IB-MATHAI-2.2-1ANSWER	IB-MATHAI-2.1-4ANSWER
Mathematics: AI 2.3 Exponential and logarithmic models	Mathematics: AI 2.2 Quadratic models	Mathematics: AI 2.2 Quadratic models
$k>0$ gives growth (N increases over time); $k<0$ gives decay (N decreases towards 0).	$x^2-5x+6=(x-2)(x-3)=0$, so $x=2$ or $x=3$.	C. In vertex form with $a<0$, the maximum value is $k=9$, occurring at $x=4$.
IB-MATHAI-2.3-1ANSWER	IB-MATHAI-2.2-4ANSWER	IB-MATHAI-2.2-3ANSWER
Mathematics: AI 2.3 Exponential and logarithmic models	Mathematics: AI 2.3 Exponential and logarithmic models	Mathematics: AI 2.3 Exponential and logarithmic models
$V(3)=24000 \times 0.9^3 = 24000 \times 0.729 = \$17,4$	B. The growth factor $1.06 = 1+0.06$, corresponding to a 6% rate per period.	Solve $N_0 e^{(kt)} = 0.5N_0$, i.e. $e^{(kt)} = 0.5$, giving $t = \ln(0.5)/k$.
IB-MATHAI-2.3-4ANSWER	IB-MATHAI-2.3-3ANSWER	IB-MATHAI-2.3-2ANSWER
Mathematics: AI 2.4 Cubic and other polynomial models	Mathematics: AI 2.4 Cubic and other polynomial models	Mathematics: AI 2.4 Cubic and other polynomial models
A. A root $x=r$ corresponds to a factor $(x-r)$; roots $-2, 1, 4$ give factors $(x+2)(x-1)(x-4)$.	A cubic can have up to two turning points and a changing concavity, letting it fit data whose trend increases then decreases then increases (or similar), which a quadratic (only one turning point) cannot represent.	As $x \rightarrow +\infty, y \rightarrow +\infty$; as $x \rightarrow -\infty, y \rightarrow -\infty$ (the graph falls on the left and rises on the right).
IB-MATHAI-2.4-3ANSWER	IB-MATHAI-2.4-2ANSWER	IB-MATHAI-2.4-1ANSWER
Mathematics: AI 2.5 Modelling with technology and model fitting	Mathematics: AI 2.5 Modelling with technology and model fitting	Mathematics: AI 2.4 Cubic and other polynomial models
Whether the model's shape and behaviour make sense in context, including under extrapolation — a model with a high R^2 over the observed range may still behave unrealistically outside it.	The proportion of the variation in the dependent variable explained by the model; R^2 closer to 1 indicates a better fit.	When the side length is 2 cm, the volume of the object is 40 cm^3 .
IB-MATHAI-2.5-2ANSWER	IB-MATHAI-2.5-1ANSWER	IB-MATHAI-2.4-4ANSWER



Mathematics: AI 2.6 Optimisation of models (HL)	Mathematics: AI 2.5 Modelling with technology and model fitting	Mathematics: AI 2.5 Modelling with technology and model fitting
Interior stationary points (where $f'(x)=0$), and the endpoints of the domain — the global maximum/minimum may occur at either.	$R^2=0.36$ is low, so the linear model explains only 36% of the variability — a poor fit, and another model type should be considered.	B. A higher R^2 (closer to 1) means the model explains more of the variability in the data.
IB-MATHAI-2.6-1ANSWER	IB-MATHAI-2.5-4ANSWER	IB-MATHAI-2.5-3ANSWER
Mathematics: AI 2.6 Optimisation of models (HL)	Mathematics: AI 2.6 Optimisation of models (HL)	Mathematics: AI 2.6 Optimisation of models (HL)
If the sides are x and $(20-x)$, $A(x)=x(20-x)=20x-x^2$, with domain $0<x<20$.	B. $f''(1)=6>0$, so $x=1$ is a local minimum.	It allows one variable to be expressed in terms of the other, reducing the problem to optimising a function of a single variable.
IB-MATHAI-2.6-4ANSWER	IB-MATHAI-2.6-3ANSWER	IB-MATHAI-2.6-2ANSWER
Mathematics: AI 3.1 Coordinate geometry and Voronoi diagrams	Mathematics: AI 3.1 Coordinate geometry and Voronoi diagrams	Mathematics: AI 3.1 Coordinate geometry and Voronoi diagrams
A. Midpoint = $((-2+4)/2, (5+(-1))/2) = (1,2)$.	That point is equidistant from all three corresponding sites — it lies on the perpendicular bisector of each pair of those sites.	It is the line perpendicular to the segment through its midpoint; every point on it is equidistant from the segment's two endpoints.
IB-MATHAI-3.1-3ANSWER	IB-MATHAI-3.1-2ANSWER	IB-MATHAI-3.1-1ANSWER
Mathematics: AI 3.2 Trigonometry: non-right triangles	Mathematics: AI 3.2 Trigonometry: non-right triangles	Mathematics: AI 3.1 Coordinate geometry and Voronoi diagrams
Because $\sin(\theta)=\sin(180^\circ-\theta)$, both an acute and an obtuse angle can give the same sine value; both should be checked against the given triangle's constraints to see whether one or both are valid.	$a^2 = b^2 + c^2 - 2bc \cdot \cos(A)$.	Distance = $\sqrt{((4-1)^2+(6-2)^2)} = \sqrt{9+16} = \sqrt{25} = 5$.
IB-MATHAI-3.2-2ANSWER	IB-MATHAI-3.2-1ANSWER	IB-MATHAI-3.1-4ANSWER
Mathematics: AI 3.3 Volume, surface area and geometric applications	Mathematics: AI 3.2 Trigonometry: non-right triangles	Mathematics: AI 3.2 Trigonometry: non-right triangles
$V = (1/3)\pi r^2 h$.	Area = $\frac{1}{2} \times 6 \times 9 \times \sin 40^\circ = 27 \times 0.6428 \approx 17.4$ (3 sf).	B. $a^2 = 8^2 + 10^2 - 2(8)(10)\cos 60^\circ = 164 - 80 = 84$.
IB-MATHAI-3.3-1ANSWER	IB-MATHAI-3.2-4ANSWER	IB-MATHAI-3.2-3ANSWER
Mathematics: AI 3.3 Volume, surface area and geometric applications	Mathematics: AI 3.3 Volume, surface area and geometric applications	Mathematics: AI 3.3 Volume, surface area and geometric applications
$V = \pi r^2 h = \pi(4)(9) = 36\pi \text{ cm}^3$.	B. Surface area = $4\pi r^2 = 4\pi(36) = 144\pi \text{ cm}^2$.	Subtract the volume of the removed solid from the volume of the original larger solid.
IB-MATHAI-3.3-4ANSWER	IB-MATHAI-3.3-3ANSWER	IB-MATHAI-3.3-2ANSWER
Mathematics: AI 3.4 Vectors (HL)	Mathematics: AI 3.4 Vectors (HL)	Mathematics: AI 3.4 Vectors (HL)
A. $2b=(4,2,-2)$; $a+2b=(1+4, 0+2, 2-2)=(5,2,0)$.	The two vectors are perpendicular (orthogonal) to each other.	$ v = \sqrt{x^2 + y^2 + z^2}$.
IB-MATHAI-3.4-3ANSWER	IB-MATHAI-3.4-2ANSWER	IB-MATHAI-3.4-1ANSWER



Mathematics: AI 3.5 Vector applications to kinematics (HL) Speed = $ v(t) $, the magnitude of the velocity vector; for $v=(v_x,v_y)$, speed = $\sqrt{(v_x^2+v_y^2)}$. IB-MATHAI-3.5-2ANSWER	Mathematics: AI 3.5 Vector applications to kinematics (HL) Velocity $v(t)=dr/dt$ (differentiate each component with respect to t); acceleration $a(t)=dv/dt$ (differentiate velocity's components again). IB-MATHAI-3.5-1ANSWER	Mathematics: AI 3.4 Vectors (HL) $ v = \sqrt{(36+64+0)} = \sqrt{100} = 10$. IB-MATHAI-3.4-4ANSWER
Mathematics: AI 4.1 Sampling techniques Proportionally to the stratum's size relative to the total population: sample from stratum = (stratum size/population size) $\times$ total sample size. IB-MATHAI-4.1-1ANSWER	Mathematics: AI 3.5 Vector applications to kinematics (HL) $t = 4-t \rightarrow 2t = 4 \rightarrow t = 2$. IB-MATHAI-3.5-4ANSWER	Mathematics: AI 3.5 Vector applications to kinematics (HL) B. $v(t)=(2t,3)$; at $t=2$, $v=(4,3)$. IB-MATHAI-3.5-3ANSWER
Mathematics: AI 4.1 Sampling techniques Systematic sampling. IB-MATHAI-4.1-4ANSWER	Mathematics: AI 4.1 Sampling techniques B. $40 \times (200/500) = 16$. IB-MATHAI-4.1-3ANSWER	Mathematics: AI 4.1 Sampling techniques It is likely to produce a biased sample, since it selects whichever individuals are easiest to access rather than representing the population fairly. IB-MATHAI-4.1-2ANSWER
Mathematics: AI 4.2 Summarising data: measures and displays C. $IQR = Q3 - Q1 = 14 - 5 = 9$. IB-MATHAI-4.2-3ANSWER	Mathematics: AI 4.2 Summarising data: measures and displays $IQR = Q3 - Q1$; it measures the spread of the middle 50% of the data and is resistant to outliers. IB-MATHAI-4.2-2ANSWER	Mathematics: AI 4.2 Summarising data: measures and displays The mean is affected by every value including outliers and can be pulled strongly towards them; the median depends only on the middle value(s) of the ordered data and is much less affected by outliers. IB-MATHAI-4.2-1ANSWER
Mathematics: AI 4.3 Bivariate data and correlation A strong correlation does not prove that one variable causes the other — the association could be due to a confounding variable, coincidence, or reverse causation. IB-MATHAI-4.3-2ANSWER	Mathematics: AI 4.3 Bivariate data and correlation $-1 \leq r \leq 1$. r near +1: strong positive linear correlation; r near -1: strong negative linear correlation; r near 0: little to no linear correlation. IB-MATHAI-4.3-1ANSWER	Mathematics: AI 4.2 Summarising data: measures and displays Median (middle of 7 ordered values) = 9. Range = $15-4 = 11$. IB-MATHAI-4.2-4ANSWER
Mathematics: AI 4.4 Probability, Venn and tree diagrams $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. IB-MATHAI-4.4-1ANSWER	Mathematics: AI 4.3 Bivariate data and correlation A moderate to fairly strong positive linear correlation between the two variables. IB-MATHAI-4.3-4ANSWER	Mathematics: AI 4.3 Bivariate data and correlation B. $x=50$ lies outside the range of the original data, so this is extrapolation, which carries a greater risk of an unreliable prediction. IB-MATHAI-4.3-3ANSWER



Mathematics: AI 4.4 Probability, Venn and tree diagrams	Mathematics: AI 4.4 Probability, Venn and tree diagrams	Mathematics: AI 4.4 Probability, Venn and tree diagrams
Outcomes greater than 4: {5,6}, so $P = 2/6 = 1/3$.	B. Mutually exclusive events cannot overlap, so $P(A \cup B) = P(A) + P(B) = 0.3 + 0.45 = 0.75$.	For independent events, $P(A \cap B) = P(A) \times P(B)$. For dependent events, $P(A \cap B) = P(A) \times P(B A)$, where $P(B A)$ is the conditional probability of B given A.
IB-MATHAI-4.4-4ANSWER	IB-MATHAI-4.4-3ANSWER	IB-MATHAI-4.4-2ANSWER
Mathematics: AI 4.5 Discrete distributions and the binomial distribution	Mathematics: AI 4.5 Discrete distributions and the binomial distribution	Mathematics: AI 4.5 Discrete distributions and the binomial distribution
B. $\text{Var}(X) = np(1-p) = 8 \times 0.3 \times 0.7 = 1.68$.	$E(X) = np$; $\text{Var}(X) = np(1-p)$.	There must be a fixed number of independent trials n, each trial has exactly two outcomes (success/failure), and the probability of success p is constant across all trials.
IB-MATHAI-4.5-3ANSWER	IB-MATHAI-4.5-2ANSWER	IB-MATHAI-4.5-1ANSWER
Mathematics: AI 4.6 The normal distribution	Mathematics: AI 4.6 The normal distribution	Mathematics: AI 4.5 Discrete distributions and the binomial distribution
$z = (x - \mu)/\sigma$; it represents the number of standard deviations x is above (positive) or below (negative) the mean μ .	Approximately 95%.	$P(X=0) = C(5,0)(0.4)^0(0.6)^5 = 0.6^5 \approx 0.0778$.
IB-MATHAI-4.6-2ANSWER	IB-MATHAI-4.6-1ANSWER	IB-MATHAI-4.5-4ANSWER
Mathematics: AI 4.7 The chi-squared test	Mathematics: AI 4.6 The normal distribution	Mathematics: AI 4.6 The normal distribution
$\chi^2 = \sum (O - E)^2 / E$, where for each cell $E = (\text{row total} \times \text{column total}) / \text{grand total}$.	Since the mean is 30, $P(X < 30) = 0.5$ by symmetry.	B. $z = (70 - 60) / \sqrt{25} = 10/5 = 2$.
IB-MATHAI-4.7-1ANSWER	IB-MATHAI-4.6-4ANSWER	IB-MATHAI-4.6-3ANSWER
Mathematics: AI 4.7 The chi-squared test	Mathematics: AI 4.7 The chi-squared test	Mathematics: AI 4.7 The chi-squared test
$df = (2 - 1)(2 - 1) = 1$.	B. A goodness-of-fit test checks how well observed data matches frequencies predicted by a hypothesised distribution.	If the calculated χ^2 statistic is greater than the critical value (from the χ^2 table at the given significance level and degrees of freedom), reject H_0 ; otherwise, do not reject H_0 .
IB-MATHAI-4.7-4ANSWER	IB-MATHAI-4.7-3ANSWER	IB-MATHAI-4.7-2ANSWER
Mathematics: AI 4.8 Spearman's rank correlation	Mathematics: AI 4.8 Spearman's rank correlation	Mathematics: AI 4.8 Spearman's rank correlation
B. r_s close to -1 indicates a strong tendency for one ranking to decrease as the other increases.	It only requires ordinal (ranked) data, detects any monotonic (not just linear) relationship, and is less affected by outliers than Pearson's r.	$r_s = 1 - 6\sum d^2 / (n(n^2 - 1))$.
IB-MATHAI-4.8-3ANSWER	IB-MATHAI-4.8-2ANSWER	IB-MATHAI-4.8-1ANSWER



Mathematics: AI 4.9 Further hypothesis tests: the t-test (HL)	Mathematics: AI 4.9 Further hypothesis tests: the t-test (HL)	Mathematics: AI 4.8 Spearman's rank correlation
If $p\text{-value} < \alpha$, reject H_0 (significant evidence against H_0); if $p\text{-value} \geq \alpha$, do not reject H_0 (insufficient evidence against H_0).	A t-test is used when the population standard deviation is unknown and must be estimated from the sample (especially with small samples); a z-test is used when the population standard deviation is known.	$\Sigma d^2 = 1 + 1 + 1 + 1 = 4$. $r_s = 1 - 6(4)/(4 \times 15) = 1 - 24/60 = 0.6$.
IB-MATHAI-4.9-2ANSWER	IB-MATHAI-4.9-1ANSWER	IB-MATHAI-4.8-4ANSWER
Mathematics: AI 4.10 Non-linear regression and the Poisson distribution (HL)	Mathematics: AI 4.9 Further hypothesis tests: the t-test (HL)	Mathematics: AI 4.9 Further hypothesis tests: the t-test (HL)
$P(X=x) = e^{(-m)} \cdot m^x / x!$, for $x=0,1,2,\dots$; $E(X) = \text{Var}(X) = m$.	Since $p=0.08 > 0.05$, do not reject H_0 — there is insufficient evidence against the null hypothesis.	B. A two-sample t-test compares the means of two independent groups/samples.
IB-MATHAI-4.10-1ANSWER	IB-MATHAI-4.9-4ANSWER	IB-MATHAI-4.9-3ANSWER
Mathematics: AI 4.10 Non-linear regression and the Poisson distribution (HL)	Mathematics: AI 4.10 Non-linear regression and the Poisson distribution (HL)	Mathematics: AI 4.10 Non-linear regression and the Poisson distribution (HL)
It suggests the power model explains about 99% of the variability in the data — an excellent fit.	B. $P(X=1) = e^{-(2^1)} / 1! = 2e^{-2}$.	Events occur independently, at a constant average rate, within a fixed interval of time or space, and two events cannot occur at exactly the same instant.
IB-MATHAI-4.10-4ANSWER	IB-MATHAI-4.10-3ANSWER	IB-MATHAI-4.10-2ANSWER
Mathematics: AI 5.1 Rates of change and the derivative	Mathematics: AI 5.1 Rates of change and the derivative	Mathematics: AI 5.1 Rates of change and the derivative
A. $d/dx(3\sin x) = 3\cos x$; $d/dx(\cos x) = -\sin x$; so $dy/dx = 3\cos x - \sin x$.	$d/dx(e^{(kx)}) = k \cdot e^{(kx)}$; $d/dx(\ln x) = 1/x$ (for $x > 0$).	Geometrically, the gradient of the tangent to $y=f(x)$ at $x=a$; physically (if f represents a quantity over time), the instantaneous rate of change of f at that point.
IB-MATHAI-5.1-3ANSWER	IB-MATHAI-5.1-2ANSWER	IB-MATHAI-5.1-1ANSWER
Mathematics: AI 5.2 Optimisation with technology	Mathematics: AI 5.2 Optimisation with technology	Mathematics: AI 5.1 Rates of change and the derivative
It allows one variable to be expressed in terms of the other, reducing the problem to optimising a function of a single variable.	That the value lies within the valid domain for the context (e.g. lengths must be positive), and that it is genuinely a maximum or minimum (via the second derivative test or a sign change of $f'(x)$).	$y' = 2x - 4$; at $x=1$, gradient $= 2(1) - 4 = -2$.
IB-MATHAI-5.2-2ANSWER	IB-MATHAI-5.2-1ANSWER	IB-MATHAI-5.1-4ANSWER
Mathematics: AI 5.3 Integration and the trapezoidal rule	Mathematics: AI 5.2 Optimisation with technology	Mathematics: AI 5.2 Optimisation with technology
$\int [a,b] f(x) dx \approx (h/2)[y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1})]$.	$A'(x) = 30 - 2x = 0 \rightarrow x = 15$ (and $A''(x) = -2 < 0$ confirms a maximum).	B. $S''(3) > 0$ confirms $r=3$ is a local minimum of $S(r)$.
IB-MATHAI-5.3-1ANSWER	IB-MATHAI-5.2-4ANSWER	IB-MATHAI-5.2-3ANSWER
Mathematics: AI 5.3 Integration and the trapezoidal rule	Mathematics: AI 5.3 Integration and the trapezoidal rule	Mathematics: AI 5.3 Integration and the trapezoidal rule
$h=1$; $(1/2)[3+4+2(5)] = (1/2)(17) = 8.5$.	A. $\int 4x^3 dx = x^4$; $\int -2x dx = -x^2$; result $x^4 - x^2 + c$.	The trapezoidal rule underestimates the true value, because straight-line chords connecting points on a concave-down curve lie below the curve.
IB-MATHAI-5.3-4ANSWER	IB-MATHAI-5.3-3ANSWER	IB-MATHAI-5.3-2ANSWER



Mathematics: AI 5.4 Kinematics using calculus	Mathematics: AI 5.4 Kinematics using calculus	Mathematics: AI 5.4 Kinematics using calculus
B. $v(t)=3t^2-3$; $a(t)=v'(t)=6t$.	Displacement is the net change in position (positive and negative contributions cancel); total distance travelled sums the absolute value of the displacement over each sub-interval of constant velocity sign, so direction reversals add rather than cancel.	$v(t)=ds/dt$ (differentiate); $a(t)=dv/dt$ (differentiate again); conversely, $s(t)=\int v(t)dt$ (integrate), using an initial condition to find the constant of integration.
IB-MATHAI-5.4-3 ANSWER	IB-MATHAI-5.4-2 ANSWER	IB-MATHAI-5.4-1 ANSWER
Mathematics: AI 5.5 Differential equations and Euler's method (HL)	Mathematics: AI 5.5 Differential equations and Euler's method (HL)	Mathematics: AI 5.4 Kinematics using calculus
Accuracy generally improves, since each linear step better approximates the curve's actual path — but more steps are then needed to reach the same x-value, increasing computation.	$y(n+1) = y_n + h \cdot f(x_n, y_n)$, and $x(n+1) = x_n + h$.	Since velocity is constant and positive, distance = displacement = $\int [0,4] 5 \, dt = 5 \times 4 = 20 \, \text{m}$.
IB-MATHAI-5.5-2 ANSWER	IB-MATHAI-5.5-1 ANSWER	IB-MATHAI-5.4-4 ANSWER
Mathematics: AI 5.6 Further calculus applications to modelling (HL)	Mathematics: AI 5.5 Differential equations and Euler's method (HL)	Mathematics: AI 5.5 Differential equations and Euler's method (HL)
$dA/dt = (dA/dr) \times (dr/dt)$ — the rate of change of A with respect to time equals the rate of change of A with respect to r, multiplied by the rate of change of r with respect to time.	$f(0,1)=1$; $y_1=1+0.1 \times 1=1.1$.	A. $f(0,0)=2(0)=0$; $y_1=y_0+h \cdot f(x_0,y_0)=0+1 \times 0=0$.
IB-MATHAI-5.6-1 ANSWER	IB-MATHAI-5.5-4 ANSWER	IB-MATHAI-5.5-3 ANSWER
Mathematics: AI 5.6 Further calculus applications to modelling (HL)	Mathematics: AI 5.6 Further calculus applications to modelling (HL)	Mathematics: AI 5.6 Further calculus applications to modelling (HL)
Area= $\int_{-2,2}(4-x^2)dx = [4x - x^3/3]$ from -2 to $2 = (8-8/3)-(-8+8/3) = 32/3$.	C. $dV/ds = 3s^2$ (the derivative of s^3 with respect to s).	Area = $\int [a,b] (f(x) - g(x)) \, dx$, integrating the difference between the upper curve and the lower curve.
IB-MATHAI-5.6-4 ANSWER	IB-MATHAI-5.6-3 ANSWER	IB-MATHAI-5.6-2 ANSWER